

HET BHALANI

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Summary

AI Engineer specializing in **Generative AI, LLMs, and Agentic Workflows**, with a strong foundation in **machine learning, deep learning, and transformer architectures**. Experienced in building **agentic workflows, REST APIs, and LLM-powered systems** using LangChain, LangGraph, and FastAPI, with hands-on expertise in **fine-tuning open-source models (QLoRA/GGUF) and end-to-end cloud-native deployment**.

Technical Skills

- **Programming Languages:** Python, JavaScript, Java, SQL, C
- **AI / ML / DL:** PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, LLM Fine-tuning (QLoRA, GGUF), Model Quantization, Transfer Learning
- **Generative AI / LLM:** LangChain, LangGraph, Groq API, RAG, Prompt Engineering, Agentic Workflows, FAISS, Pinecone, LangSmith, LangFuse
- **MLOps / Cloud / Web:** Docker, CI/CD, Apache Airflow, AWS (EC2, S3, IAM), Redis, FastAPI, Next.js, REST API, Git, Linux

Projects

HetGPT - Full-Stack AI Assistant

Dec 2025 - Feb 2026

Python, FastAPI, LangChain, QLoRA, GGUF, FAISS, Pinecone, Docker, Next.js

- Developed a generative AI assistant with a **dual memory architecture** (short-term context + long-term summarization), **document RAG** (FAISS/Pinecone vector search), and **tool-use integrations** for real-time web search and weather.
- Fine-tuned **Qwen3-8B via QLoRA**, quantized to **Q4_K_M GGUF**, and deployed as a **REST API on Hugging Face Spaces**, achieving **fast inference speeds** via a llama.cpp Docker backend; secured with **JWT authentication** and device-level rate limiting.

BuzzBot - Production-Grade AI Content Pipeline

Apr 2026 - May 2026

Python, LangGraph, Apache Airflow, Docker, AWS (EC2, S3, IAM), Tavily, Groq, Redis, LangSmith, LangFuse

- Built an autonomous agentic pipeline using **LangGraph** and **Apache Airflow** that fetches, deduplicates, and reranks **100+ daily news articles**, persisting structured data securely into **AWS S3** using scoped IAM roles.
- Designed a **non-blocking Human-in-the-Loop (HITL)** workflow with a **Telegram Bot** and **Redis checkpointing**, suspending processes via `interrupt()` for manual review before auto-publishing to LinkedIn, saving **5+ hours per week** of manual curation.
- Deployed the multi-container architecture on **AWS EC2** via **Docker** and **GitHub Actions CI/CD**, integrating **MLflow/LangFuse** to systematically version prompts and track generation quality metrics.

Lexi - AI Writing Assistant (Chrome Extension)

Feb 2026 - Mar 2026

Python, FastAPI, LangGraph, Groq API, Chrome Extension MV3, MyMemory API

- Developed a **Chrome Extension (MV3)** with a **LangGraph-orchestrated FastAPI backend** supporting grammar correction, summarization, tone control, and a session-based **iterative rewrite workflow** (up to 3 rewrites) serving dynamic web users.
- Engineered **multilingual translation capabilities across 15+ languages** with automatic language detection and a **Promptify feature** to convert raw user intents into structured, high-yield LLM prompts.

Experience

Teaching Assistant - Data Structures and Algorithms

Dec 2024 - May 2025

Darshan University, Rajkot, Gujarat

- Mentored **100+ junior students** across **80+ lab sessions** in data structures and algorithms, guiding hands-on **DSA problem-solving and implementation** through code reviews and one-on-one teaching.

Education

Bachelor of Technology (B.Tech) - Computer Science and Engineering

Expected May 2027

Darshan University, Rajkot, Gujarat

CGPA: 8.97 / 10.0

Achievements

- **Neural Network from Scratch** gained **37 GitHub stars and 8 forks** organically, demonstrating a strong foundation in deep learning math and backpropagation.
- Led a **team of 6** in a **24-hour hackathon** (Code Arena), successfully architecting and delivering an **ML-powered finance optimizer** with personalized recommendations under strict time constraints.